

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

May 2012

CL310 Irrigation & Hydraulic Structures

Date: 11.05.2012, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Explain "Irrigation". What are the advantages and disadvantages of irrigation? [04]
- (b) Farmers are given subsidy, if they practice drip irrigation method. Justify the statement with proper arguments. [03]
- Q - 2 (a) An outlet is to irrigate 4800 ha. of the land in rabi season and 1600 ha. in kharif season. Kor depth and kor period for rabi season are 14 cm and 24 days respectively while those for Kharif season are 20cm and 16days. Determine the design discharge of an outlet. [06]
- (b) Explain with a neat sketch, frequency of irrigation. What are the factors affecting the frequency of irrigation? Discuss any one of them in detail. [05]
- (c) Explain the necessity of canal fall in a canal network. What are the different types of canal falls which can be used as discharge measurement? [03]

OR

- Q - 2 (a) Determine the design discharge of the canal for the cropping pattern given in the table for an irrigation command area of 50000 ha, out of which 10% is uncultivable. [06]

Crop	Base period (days)	Intensity of irrigation (%)	Duty at the head (ha/cumecs)
Sugarcane	365	25	650
Paddy(Kharif)	115	40	750
Bajra (Kharif)	110	20	1200
Wheat	110	45	1000
Vegetable(HW)	75	35	650

Take time factor = 0.7, capacity factor = 0.8 and conveyance efficiency= 75 %

- (b) Define: [05]
Command area, Delta, Intensity of irrigation, Field capacity, Kor period
- (c) Distinguish between Aqueduct and Super passage. Sketch the cross-section of an aqueduct. [03]

- Q - 3 (a) Enumerate the classification of cross drainage works. [02]
- (b) Draw the diagram showing layout of diversion head works and state function of each component. [06]
- (c) Describe in brief various types of weirs. Distinguish clearly between a weir and a barrage. [06]

OR

- Q - 3 (a) Enlist factors affecting selection of type of cross drainage works. [02]
- (b) What are the causes of the failure of weir founded on permeable soils? Suggest the remedial measures for the same. [06]
- (c) Define creep length. Explain Bligh's creep theory for the design of impervious floor on permeable soils. Also, discuss the limitations of this theory. [06]

SECTION - II

- Q - 4 (a) Enlist types of canals. [02]
- (b) Using Kennedy's theory, design a channel section using the following data: [05]
Discharge, $Q=25$ cumec, Kutter's $N=0.0225$, Critical velocity ratio, $m=1$
Side slope $=0.5:1$ and Bed slope, $S=1$ in 5000
- Q - 5 (a) Design and draw a low gravity dam having following data: [07]
Maximum allowable compressive stress in concrete $= 37.5 \text{ kg/cm}^2$
Unit weight of water $= 1000 \text{ kg/m}^3$
Specific weight of concrete $= 2.4$
Assume freeboard $= 2.3 \text{ m}$
- (b) State and explain the causes of failure of structure founded on permeable soil with neat sketches. [07]

OR

- Q - 5 (a) Figure 1 shows the section of a gravity dam. Calculate the water pressure, self weight and uplift pressure of a dam with detail pressure distribution diagrams of each one. [07]

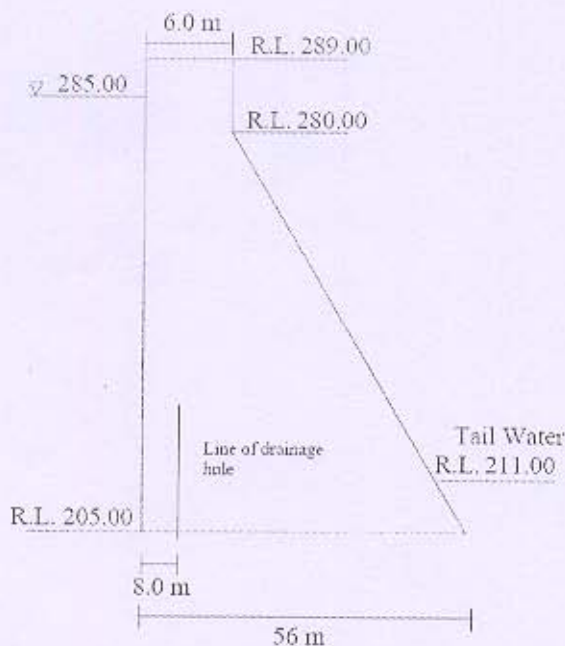


Figure 1

- (b) Describe with neat sketches, the various methods adopted for controlling seepage through the body of the dam and through foundation. [07]
- Q - 6 (a) Draw a neat sketch showing the cross section of a rock-fill dam. [02]
- (b) With neat sketch discuss the ogee spillway. [05]
- (c) Draw neat sketches of followings showing all the components: [07]
- Straight drop spillway without d/s protection
 - Straight drop spillway with d/s protection
 - Shaft spillway

OR

- Q - 6 (a) Draw a neat sketch of Chimney drain in embankment dam. [02]
- (b) Define spillway. What is the purpose to provide it? What are the essential requirements? Where the spillway is located? [05]
- (c) In which condition a hydraulic jump takes place? Differentiate Jump Height Curve (J.H.C.) and Tail Water Curve (T.W.C.) with neat sketch. Also discuss the five possibilities of hydraulic jump formation with neat diagrams. [07]
